

Date: 09/15/26

Time: 6:00 - 6:20

Location: Virtual meeting

Team members present: Joseph, Alexander, Tram, Umair Waleed, Rahul Nesan, Niles, Dhruvi Dharmeshbhai

Tasks allocated/completion:

- **Initial UI and cell states** (estimated completion time: 1 hour) ✓
 - Create a minesweeper grid and label columns and rows
 - Create cell states like **cell.is_mine**, **cell.is_uncovered**, and **cell.is_flagged**, and the cell counter called **cell.adjacent_mines**

Author: Joseph, Date: 9/15, Actual completion time: 1 hour

- **Game status and mines remaining** (estimated completion time: 1 hour) ✓
 - Print the player's status while playing the game. Change upon uncovering a mine, or uncovering all cells without a mine
 - Print the remaining amount of mines. Referring to the number of flags placed on a cell that could or could not have mines.

Author: Joseph, Date: 9/15, Actual completion time: 1.5 hours

- **Mine configuration** (estimated completion time: 1.5 hrs) ✓
 - Prompt the player for the desired mine count (10-20)
 - Randomly place the mines on the grid

Author: Dhruvi, Date: 9/16, Actual completion time: 50 min

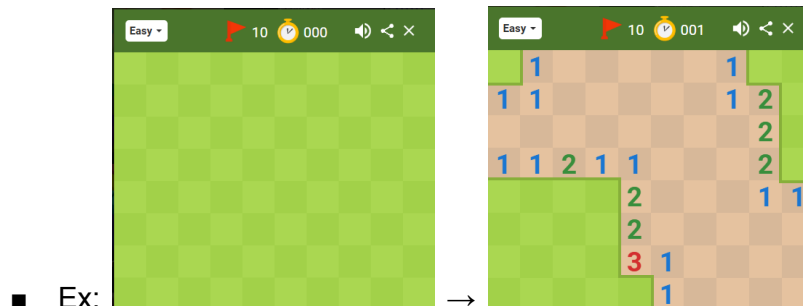
- **Adjacent mine counter** (estimated completion time: 2 hrs) ✓
 - Update the **cell.adjacent_mines** for each cell near a **cell.is_mine** cell. Try adding a function within the mine configuration code and select the adjacent 2D indices to increase the **cell.adjacent_mines** value.

Author: tram dinh, Date: 09/16, Actual completion time: 2 hours and 40 mins

- Add Uncover function and Main loop (estimated completion time: 1.5 hrs) ✓
 - Create the uncover_cell function for the player and for other functions to use
 - Add the main loop and end the program once the player wins or loses

Author: Rahul Nesan Jawahar Nesan: , Date:09/17, Actual completion time: 2 hrs

- Add the player's first safe-uncovered cell and the auto-uncovered adjacent cells (estimated completion time: 1.5) ✓
 - Add the player's first safe-uncovered cell (so place the mines after the player's first selected cell to uncover)
 - Add the feature where the game uncovers adjacent cells. Check if the adjacent cell has the **cell.adjacent_mines** value as zero; if so, uncover until you encounter a neighboring cell with a **cell.adjacent_mines** value not equal to zero.



Author: Alexander Wichern, Date: 9/17, Actual completion time: 1.5 hours

- Add Flag function (estimated completion time: 2 hours) ✓
 - Create the flag function for the user
 - Allow the same flag function to unflag a cell
 - Prevent uncovering a cell that is currently flagged

Author: Umair Chaudhry, Date: 9/17, Actual completion time: 1 hour

- Add player input (estimated completion time: 2 hours) ✓
 - Define an input string format for the player
 - Ex. A4, Flag **OR** Flag, A4
 - Format and run the player's command
 - Show the mine locations when the player loses

Author: Niles Chen, Date: 9/18, Actual completion time: 3 hours

Person-Hours Estimate methodology:**Unadjusted Use Case Weight (UUCW):**

(Simple ≤ 3 steps = 5 pts, Average 4–7 = 10 pts, Complex > 7 = 15 pts)

Unadjusted Actor Weight (UAW):

(Simple = 1 pts, Average = 2 pts, Complex = 3 pts)

Unadjusted UCP (UUCP):

(UUCW + UAW)

Task	Steps = UUCW	Actor = UAW	UUCP
Initial UI & cell states	2 steps = Simple (5)	System (1)	6
Game status & mines remaining	2 steps = Simple (5)	System (1)	6
Mine configuration	2 steps = Simple (5)	Player (2)	7
Adjacent mine counter	Loop+bounds+increment = Average (10)	System (1)	11
Uncover function & main loop	Flood-fill + loop control = Average (10)	Player (2)	12
First-safe-move + auto-uncover	Relocate mine + recursive flood-fill = Complex (15)	Player (2)	17
Flag function	3 steps = Simple (5)	Player (2)	7
Player input parsing	Multi-format parsing + validation = Average (10)	Player (2)	12
Total			78

Technical & Environmental Factors:

TCF & ECF: $TCF = 0.6 + (0.01 \times \Sigma \text{ technical weights})$; $ECF = 1.4 + (-0.03 \times \Sigma \text{ environmental weights})$.

Factor	Description	Rating (0–5)
T1	Distributed System	0
T2	Performance Objectives	1
T3	End-user Efficiency	2
T4	Complex Internal Processing	3

T5	Reusability	1
T6	Easy to Install	5
T7	Easy to Use	3
T8	Portability	4
T9	Easy to Change	3
T10	Concurrency	0
T11	Special Security Features	0
T12	Third-party Access	0
T13	Special Training Needs	0
	Sum of technical factor weights	22

Factor	Description	Rating (0–5)
E1	Familiarity with UCP	1
E2	Application Experience	2
E3	Object-oriented Experience	3
E4	Lead Analyst Capability	3
E5	Motivation	4
E6	Stable Requirements	4
E7	Part-time Staff	5
E8	Difficulty of Language	1
	Sum of environmental factor weights	23

Technical Complexity Factor (TCF)

$$\text{TCF} = 0.6 + (0.01 \times 22) = \mathbf{0.83}$$

Environmental Complexity Factor (ECF):

$$\text{ECF} = 1.4 + (-0.03 \times 23) = \mathbf{0.71}$$

Adjusted UCP:

$$78 \times 0.83 \times 0.71 \approx 45.97$$

Productivity factor: calibrated to our actual hours — $13.5 \text{ hrs} \div 45.97 \text{ UCP} \approx 0.294 \text{ hrs/UCP}$.

Applied per task:

Task	Original Estimate	Actual	UCP-Redefined Estimate
Initial UI & cell states	1 hr	1 hr	1.0 hr
Game status & mines remaining	1 hr	1.5 hr	1.0 hr
Mine configuration	1.5 hr	0.83 hr	1.2 hr
Adjacent mine counter	2 hr	2.67 hr	1.9 hr
Uncover function & main loop	1.5 hr	2 hr	2.1 hr
First-safe-move + auto-uncover	1.5 hr	1.5 hr	2.9 hr
Flag function	2 hr	1 hr	1.2 hr
Player input parsing	2 hr	3 hr	2.1 hr
Total	12.5 hr	13.5 hr	13.5 hr

Notes: We decided to make a list of tasks to complete for the project. Each member will choose a task to complete and commit to a main branch.